

THE FILTER IS NOT THE PROOF

From Filtration Performance to Atmospheric Accountability

Technical Governance Reference | v2.0 | 10 October 2026

PRE-EXAMINATION | BASELINE NOT FROZEN | NO PERFORMANCE CERTIFICATION

This reference proposes an examination method. It is not a filtration design standard, legal opinion, product endorsement, installation certification, or independently validated performance finding.

01 Executive proposition

A filter can be correctly specified and installed without proving that the occupied space receives the intended reduction in exposure. A measured improvement may support a bounded technical conclusion, but cannot alone establish a health outcome or permission to modify equipment. TA14 separates observation, evidence, authority, execution, and verified consequence.

Governing question: Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Constitutional chain: Reality - Record - Continuity - Admissibility - Binding - Commit - Execution - Outcome.

02 Define the claim before measuring

Claim level	Typical supporting evidence	What it does not prove
Product rating	Relevant test standard and tested configuration	Installed-system performance
Installation	Identity, fit, seals, bypass inspection, commissioning	Delivered clean air in occupied zones
Operating performance	Airflow, loading, pressure, operational context, appropriate removal test	Reduced personal exposure or health effect
Indoor-air change	Validated concentration comparisons, location/time, sources, confounders	Attribution to filter alone
Human outcome	Appropriate exposure and health-study design	Automatically established by a maintenance test

03 Engineering boundaries

MERV / ASHRAE 52.2: a particle-removal rating under specified laboratory test conditions; not a promise of installed whole-building performance. **HEPA:** a high-efficiency filter designation whose applicable test method and system integrity must be specified. **ISO 16890:** a separate classification framework for general ventilation filters; do not treat its categories as direct MERV equivalents without a documented method.

Pressure differential: useful for diagnosing loading and operating state, but affected by airflow and system configuration; it does not directly measure particle removal. **Airflow and bypass:** fan capacity, leaks, poor seating and unfiltered paths may dominate installed performance. **Portable cleaners:** distinguish product clean-air delivery rating,

placement, operating speed, noise-related use, room mixing and actual runtime.

04 Eight-stage examination

01 PROMISE: Define target contaminant, performance criterion, operating envelope, responsible party and claim type.

02 INSTALLATION: Establish filter identity, correct fit, seal condition, orientation, bypass, commissioning and deviations.

03 OPERATING REALITY: Record actual airflow, fan state, differential pressure, occupancy, sources and indoor/outdoor conditions.

04 EVIDENCE QUALITY: Preserve raw readings, method, device identity, calibration/verification, location, time and uncertainty.

05 CONTINUITY: Link replacements, alarms, faults, changes and superseded records; retain a visible correction trail.

06 AUTHORITY: Establish who can approve the work, under what contract, code, safety rules and delegated scope.

07 EXECUTION: Capture approved scope, performer, work order, exceptions, before/after state and safety checks.

08 OUTCOME: Verify the specified technical result and state explicitly what remains unproven.

05 Four failure modes for examination

Bypass despite a high-rated filter. Installation evidence and seal inspection are required before inferring delivered performance.

Loaded filter with reduced airflow. A high pressure differential prompts diagnosis; it does not automatically authorize replacement or prove reduced protection.

Drifting or misplaced sensor. Flag validity, preserve original data, record corrections and avoid silently promoting readings to fact.

Unauthorized control change. Even a technically plausible response may require HOLD or ESCALATE until local authority and safety impact are established.

06 Illustrative Atmospheric Integrity Record

Field	Illustrative entry / required handling
Asset and claim	AHU-01; replace filter if defined criteria and authorized maintenance process are satisfied
Observation	Elevated differential pressure at recorded fan speed; raw value and timestamp retained
Provenance	Instrument ID, verification date, sampling point, technician, method and uncertainty
Context	Fan speed, airflow evidence, occupancy, outdoor particle conditions, bypass inspection
Authority	Owner-approved maintenance scope; responsible supervisor; applicable safety constraints
Disposition	HOLD pending verification of airflow and measurement; then reassess
Execution / outcome	If authorized: replacement record, seal check, post-work readings and bounded finding
Correction history	Append amendment with reason, author and timestamp; retain original observation

All values above are **illustrative placeholders**, not records from a real installation. Raw and adjusted values must remain separately identifiable, with method and uncertainty documented.

07 Decision protocol: ALLOW / HOLD / DENY / ESCALATE

ALLOW: evidence is sufficient for the bounded proposed task and local authority/standing are established. **HOLD:** a measurement, baseline, criterion or approval remains unresolved. **DENY:** the proposed action is outside an applicable prohibition or cannot meet required conditions. **ESCALATE:** legal, safety, jurisdictional or specialist determination exceeds delegated authority. Revalidate after material context changes.

08 Owner liability, enforcement and record integrity

Owner liability: records may be discoverable. Retention, privacy, preservation duties and correction rules require qualified legal review; TA14 promises no immunity or safe harbor. **Enforcement:** structured evidence can assist review but cannot create staff, jurisdiction or expertise. **Measurement integrity:** a recorded error remains an error; record provenance, validity flags, versioned corrections and dissenting assessments rather than overwriting history.

09 Clean-air pledges and regulatory duties

A voluntary pledge, product rating, contractual commitment and legal obligation are different authorities. Establish the exact issuer, text, selected commitments, relevant jurisdiction and dutyholder before assessing fulfillment. A promise to use filtration is not proof that the desired indoor-air condition was achieved; a technical result is not automatically proof of a health outcome.

10 Proposed pilot / independent examination

Before field examination: (1) name the building, asset and scope; (2) freeze a versioned baseline; (3) define target claim and acceptance criteria; (4) record measurement method and uncertainty; (5) identify decision-maker and authority basis; (6) predefine exception and correction handling; (7) record authorized intervention; (8) verify outcomes over a defined window; (9) publish bounded findings, failures and unresolved questions.

11 Showroom teaching sequence

Image 01 Promise; 02 Installation; 03 Operating Reality; 04 Evidence Quality; 05 Continuity; 06 Authority; 07 Execution; 08 Outcome. Each image should teach the specific boundary, with numbered narration and no implied certification.

12 Institutional independence and invitation

This is a TA14-authored proposal for critical examination. No filtration specialist, manufacturer, school, regulator, pledge organization or other party is represented as a co-author or endorser. Filtration practitioners are invited to challenge assumptions, test failure modes and help define measurable, bounded demonstrations. No formal examination disposition is issued by this reference.

Observed is not inferred. Inferred is not authorized. | Capability is not authority.